

Consider the repeating decimal $0.5\bar{8}$ for questions 1-4.

1. If $x = 0.5\bar{8}$, complete each equation.

$$x = 0.5\bar{8}$$

$$10x = \underline{5.\bar{8}}$$

$$100x = \underline{58.\bar{8}}$$

2. When subtracted, which two values would result in the repeating portion of the decimal being eliminated?

$$58.\bar{8} - 5.\bar{8}$$

3. Use the two relationships represented in question 1 to complete the steps to find the fraction equivalent of $0.5\bar{8}$.

$$\underline{100x} - \underline{10x} = \underline{58.\bar{8}} - \underline{5.\bar{8}}$$

$$\underline{90}x = \underline{53}$$

$$x = \frac{\boxed{53}}{\boxed{90}}$$

4. Write the fraction and mixed number that is equivalent to $3.5\bar{8}$.

$$3\frac{53}{90} = \frac{323}{90}$$

A trail mix recipe calls for the following ratios: $21.\overline{25}\%$ raisins, $\frac{5}{9}$ peanuts, 0.14 chocolate chips, and the rest is made up of chex mix.

5. What fraction of the recipe is made up of chocolate chips?

$$0.14 = \frac{14}{100} = \frac{7}{50}$$

6. What fraction of the recipe is made up of raisins?

$$\begin{aligned} 21.\overline{25}\% &= 0.21\overline{25} \\ 100x - x &= 21.\overline{25} - 0.21\overline{25} \\ 99x &= 21.04 \\ x &= \frac{21.04}{99} = \frac{2,104}{9,900} = \frac{526}{2,475} \end{aligned}$$

7. What is the decimal equivalent of the peanuts?

$$\frac{5}{9} = 0.\overline{5}$$

8. Approximately what percent is made up of chex mix? Round your answer to the nearest whole percent.

$$1 - \left(\frac{526}{2,475} + \frac{5}{9} + \frac{7}{50} \right) \approx 0.09 = 9\%$$

For questions 9-10, complete the table below.

	Decimal	Percent	Fraction (or Mixed Number)
9.	$9.1\overline{52}$	$915.2\overline{52}\%$	$9\frac{151}{990}$
10.	$0.01\overline{6}$	$1.\overline{6}\%$	$\frac{1}{60}$